

PROJECT PLAN DOCUMENT

(Due: 12th February)

Project number	2
Project Title	Personal Health Chatbot
Document	Project Plan
Creation date	11th February 2026
Created By	Venya Velmurugan, Advait Dintakurti, Vikesh Kansal, Keshav Goel, Amish Goyal
Client	NanoCare Health Services Private Limited

Brief Problem Statement

Electronic Health Records (EHRs) contain critical patient information but are often verbose, unstructured, and difficult to interpret efficiently in clinical contexts. Standard large language models lack personalization and clinical grounding, leading to generic advice that does not account for individual patient histories. For example, treatment recommendations must differ significantly for a diabetic patient versus a non-diabetic one. This project aims to develop a "clinically aware" chatbot that provides personalized medical guidance by synthesizing data from the client's existing EMR system to create a comprehensive "Medical ID". The system will integrate structured medical knowledge graphs with controlled EHR retrieval to improve diagnostic accuracy, reduce hallucination risks, and ensure clinically grounded reasoning while adhering to strict privacy constraints through data anonymization and eventual migration to self-hosted infrastructure.

Team Members

- **Venya Velmurugan** – Developing multilingual interface support and integrating UMLS/SNOMED CT knowledge graphs within the system's reasoning pipeline.
- **Advait Dintakurti** – Designing the modular system architecture and implementing the backend–frontend integration framework.
- **Vikesh Kansal** – Implementing hybrid retrieval (semantic + structural search) and medical entity mapping to standardized codes.
- **Keshav Goel** – Working on retrieval ranking strategies, evaluation mechanisms, and system validation testing.

- **Amish Goyal** – Developing EMR ingestion, data processing modules, and core backend pipeline components.

Team Communication

The team conducts weekly structured meetings with the client mentor (NanoCare Health Services Private Limited) every Wednesday at 4:00 pm to review progress, gather feedback, and align on priorities. Internal team coordination meetings occur at least twice weekly for technical synchronization and task allocation. Primary communication channels include WhatsApp for daily coordination and quick updates and GitHub for version control and code collaboration. Weekly status updates are maintained through a project tracker to monitor individual contributions and milestone progress.

Development Environment

- **Programming Language:** Python 3.x
- **LLM Integration:** Gemini API (current prototype phase), with modular architecture for future migration to self-hosted local LLM
- **Data Format:** JSON-based structured EMR data from client's existing system
- **Backend Framework:** Python scripts with potential migration to FastAPI for production deployment
- **Frontend:** Basic web interface (to be finalized), with consideration for Next.js or similar modern framework
- **Vector Database:** FAISS or equivalent for semantic search and embedding storage
- **Knowledge Graph:** UMLS/SNOMED CT-based structured medical knowledge representation
- **Version Control:** Git and GitHub for collaborative development
- **Documentation:** Microsoft Word, Google Docs, and Markdown files
- **Testing:** Manual testing, scenario-based clinical evaluation, and PyTest for backend validation.
- **Collaboration Tools:** WhatsApp, Google Drive, GitHub Issues for task tracking

System Architecture Overview

The system follows a four-phase pipeline architecture designed to ensure clinically grounded and personalized medical reasoning:

Phase 1: ETHER INGESTION (The Organizer)

This phase processes raw EMR files and prepares structured data for retrieval:

- **A. Relative Time Encoding:** Converts absolute timestamps to relative time offsets (e.g., "Day 5", "Day 500") to establish temporal context
- **B. Semantic Stream Split:** Generates sentence embeddings for conceptual matching, creating a vector index of clinical narratives

- **C. Structural Stream Split:** Extracts structured data elements such as lab values, medication codes, and diagnostic labels for precision keyword matching

Phase 2: INITIAL RETRIEVAL (The Dual Search)

This phase retrieves the most relevant patient history events using a hybrid approach:

- **A. Semantic Search:** Query vector matches against the vector index for conceptual relevance
- **B. Structural Search:** Query keywords match against the keyword index for exact data point retrieval
- **C. Merge & Rank (RRF):** Combines results using Reciprocal Rank Fusion to identify Top-N relevant events from patient history

Phase 3: KG CONSULTATION (The Rulebook)

This phase validates retrieved information against established medical knowledge:

- **A. Entity Mapping:** Maps Top-N events to standardized medical codes (CUIs) from UMLS or SNOMED CT
- **B. Knowledge Graph Query:** Retrieves relevant medical rules, contraindications, and diagnostic criteria connected to identified entities
- **C. Feedback Loop:** Identifies missing evidence based on KG rules and triggers targeted re-scan if needed to fill diagnostic gaps

Phase 4: FINAL SYNTHESIS (The Judge)

This phase generates the final clinical output using controlled LLM reasoning:

- **A. LLM Input Package:** Combines Top-N events timeline (evidence) with retrieved medical rules (constraints) into a structured prompt
- **B. Reasoning & Generation:** LLM applies rules to evidence, checks for contradictions, and generates clinical insights and personalized recommendations based on retrieved evidence
- **C. Output:** Delivers personalized diagnosis, treatment recommendations, and clinical explanations grounded in patient history

Key Risks and Mitigation Strategies

Risk	Mitigation Strategy
Negation Handling Errors	Implement explicit negation tagging during EMR ingestion to prevent misinterpretation (e.g., "no history of diabetes" vs "diabetes")
Temporal Relevance Drift	Apply temporal weighting during retrieval phase to prioritize recent medical events while maintaining access to historical context

Knowledge Graph Bias	Restrict KG to validation and rule-checking role rather than sole diagnosis driver; maintain LLM reasoning with evidence-first approach
Token Efficiency & Context Limits	Implement controlled Top-N retrieval with configurable threshold and gated KG expansion to manage context window constraints
LLM Hallucination Risk	Use evidence-first prompting strategy, rule-constrained generation, and explicit citation of source events in output
Privacy & PII Exposure	Implement data anonymization layer before API calls; design modular architecture for eventual migration to self-hosted local LLM
EMR Data Quality Issues	Develop robust parsing and validation logic to handle inconsistent formatting, missing fields, and unstructured clinical notes
Client Feedback Integration	Maintain flexible architecture with weekly milestone reviews to accommodate evolving requirements and feedback-driven changes
Lack of real-time location awareness	Integrate controlled external data sources (e.g., weather alerts, epidemiological updates) to provide contextual awareness of real-time conditions in the patient's location.

Table 1: Risk Assessment and Mitigation Strategies

Milestone Schedule

Milestone	Due Date	Release	Deliverable?
Fix meeting with client	Week 1 (Jan 19-25)	R1	No
Identify user requirements	Week 1 (Jan 19-25)	R1	No
Project synopsis writeup	Week 1 (Jan 19-25)	R1	Yes
Submitted to client	Week 1 (Jan 19-25)	R1	No

Client meeting	Week 2 (Jan 26-Feb 1)	R1	No
Developing technical architecture	Week 2 (Jan 26-Feb 1)	R1	No
Researched alternatives to Vanilla RAG	Week 2 (Jan 26-Feb 1)	R1	No
Develop broad timeline based on proposed architecture	Week 2 (Jan 26-Feb 1)	R1	No
Submit project plan to client	Week 2 (Jan 26-Feb 1)	R1	Yes
Client meeting	Week 3 (Feb 2-8)	R1	No
Look into RAG vs direct context	Week 3 (Feb 2-8)	R1	No
Test FHIR-RAG prompts	Week 3 (Feb 2-8)	R1	No
Research on Medical Standards (UMLS)	Week 3 (Feb 2-8)	R1	No
Look into UMLS licensing	Week 3 (Feb 2-8)	R1	No
Develop basic prototype	Week 3 (Feb 2-8)	R1	No
Analyze limitations of Knowledge Graph method	Week 3-4 (Feb 2-15)	R1	No
Client meeting	Week 4 (Feb 9-15)	R2	No
Requirements document finalized	Week 4 (Feb 9-15)	R2	Yes
Implement EMR ingestion module with relative time encoding	Week 5 (Feb 16-22)	R2	No
Develop semantic and structural retrieval pipeline	Week 5-6 (Feb 16-Mar 1)	R2	No

Integrate Reciprocal Rank Fusion (RRF) for hybrid search	Week 6 (Feb 23-Mar 1)	R2	No
Implement entity mapping to UMLS/SNOMED CUIs	Week 6-7 (Feb 23-Mar 8)	R2	No
Develop Knowledge Graph consultation layer	Week 7 (Mar 2-8)	R2	No
Backend code complete	Week 7-8 (Mar 2-15)	R2	No
Design and implement frontend interface	Week 8-9 (Mar 9-22)	R2	No
Integrate frontend with backend API	Week 9 (Mar 16-22)	R2	No
End-to-end prototype demonstration	Week 10 (Mar 23-29)	R2	Yes
Develop clinical test scenarios	Week 10-11 (Mar 23-Apr 5)	R3	No
Conduct system testing and validation	Week 11-12 (Mar 30-Apr 12)	R3	No
Evaluate hallucination prevention mechanisms	Week 12 (Apr 6-12)	R3	No
Document system architecture and codebase	Week 12-13 (Apr 6-19)	R3	No
Prepare final demonstration	Week 13 (Apr 13-19)	R3	No
Final system demo and documentation submission	Week 14 (Apr 20-26)	R3	Yes

Table 2: Project Milestone Schedule

Note: This is a living document. Milestone dates and deliverables are subject to change based on weekly client feedback, technical challenges encountered during

implementation, and evolving project requirements. The schedule will be updated regularly to reflect actual progress and adjusted priorities.